

Grep

short option	long option	description
-i	--ignore-case	ignore case distinctions
-v	--invert-match	displays lines that do not match
-c	--count	returns number of matching lines
-n	--line-number	prefixes each output line with its line number
-l	--files-with-matches	only print names of files
-r, -R	--recursive	recursively searches directories
-w	--word-regexp	matches only whole words
-o	--only-matching	print only matched text

sed → sed stands for stream editor

sed [options] 'command' filename

Options	Long Form	Description
-i	--in-place	Edit the file directly in place (overwrite the original file)
-n	--quiet, --silent	Suppress the default automatic printing of pattern space
-E / -r	--regexp-extended	Enables extended regular expressions
-e	--expression	Allow you to chain & run multiple sed expressions

cut → It is use to slice & extract specific sections from each line of file

Options	Description
-d	Defines the delimiter
-f	Specifies which fields to extract
-c	Extracts by character position

wc → counts the number of lines, words & bytes/characters in a file

Options	Description
-l	Prints only the line count
-w	Prints only the word count
-c	Prints only the byte count
-m	Prints only the character count

sort → Rearranges lines of text alphabetically & numerically

Options [1, 2, 3, 4, 5] Description

-n	Sorts numerically
-r	Reverse the sorting order
-k	Sort based on a specific column
-t	Defines the field delimiter

Error handling & Debugging

- 1) exit codes — exit 0 → success exit 1 → miscellaneous errors, general application runtime failures
- 2) set -e — it stops when error occurred
- 3) set -u — block the execution of undefined variable
- 4) set -o pipefail — catches hidden failure buried inside piped command